

# Atlas of Constants 02

— On the Speed of Light ( $c$ )

Sequential Closure,

and the Invariant Null Slope.

# 0. Atlas Header (Standardized)

Atlas ID: AoC\_2

Internal ID: Appx\_7\_AoC\_02\_On\_c\_1.2

Symbol: c

Role (Conceptual)

Sequential invariant / null-slope constraint.

This entry treats c as the structural limit of sequence, not a calibrated speed: the invariant null-slope that allows events to form a stable before/after relation across admissible frames.

Dependency Spine (Prooffield Index v2.1)

$$\text{AoC}_0(\hat{\Omega}) \rightarrow \text{AoC}_1(\pi) \rightarrow \text{AoC}_2(c) \rightarrow \text{AoC}_3(\alpha) \rightarrow \{\text{AoC}_4(m_p), \text{AoC}_5(m_n), \text{AoC}_6(G), \dots\}$$

This entry depends on:

- AoC\_00\_The\_Collapse\_Operator\_vX.X — operator, Units Firewall, viability map.
- AoC\_01\_On\_Pi\_vX.X — spatial closure / geometric survivability.

It feeds:

- AoC\_03\_On\_137\_vX.X
- AoC\_04\_On\_Proton\_Mass\_vX.X
- AoC\_05\_On\_Neutron\_Mass\_vX.X
- AoC\_06\_On\_Gravity\_vX.X

and any appendix that assumes a rigid light-cone background.

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# Epistemic Key (Tag Registry for This Entry)

All load-bearing claims in this document are explicitly labeled from the following set:

- [DERIVED] – Follows from AoC\_00, AoC\_01, and the declared Ansätze by explicit argument.
- [ANSATZ] – Structural assumption adopted for this entry (e.g., signature, admissible frames) and surfaced as such.
- [PHENOMENOLOGICAL REQUIREMENT] – Constraint taken from observed, lived universes (e.g., directional causality) rather than pure geometry.
- [CONJECTURE] – Plausible but not yet fully supported by  $\hat{\Omega}$ -based derivation or stress tests.
- [CALIBRATION] – Mapping from dimensionless survivors to unit-bearing quantities (e.g., from  $k_{\star}$  to  $c$  in SI); not part of the structural derivation.

No untagged load-bearing claims are permitted in the body of this entry.

## Collapse Consequence (Summary)

If no finite, invariant null-slope survives, sequencing fails.

Causality becomes frame- and geometry-dependent noise; “before” and “after” cannot be stably defined, and durable physical histories (including any Atlas entry that assumes ordered evolution) collapse.

This consequence is intended as falsifiable within the declared scope of this entry: any Lorentzian universe that satisfies the Phenomenological Sequence Requirement (ordered, lived causality as defined in §3.2.3) and nonetheless demonstrably maintains stable, observer-independent histories without an invariant null-slope structure would falsify AoC\_2’s necessity claim.

## AGI Verifier Hint

Start here if you are:

- Verifying causal invariance or light-cone rigidity across frames or geometries.
- Testing whether a candidate spacetime admits a finite, invariant, dimensionless null-slope  $k_{\star}$  under  $\hat{\Omega}$ ’s viability map.

- Probing information-theoretic ceilings: this entry treats  $c$  as the structural ceiling on causal influence, converting spatial coherence (AoC\_01:  $\pi$ ) into temporal order (sequencing).

This constant should be read as infrastructure for the Atlas: if this entry, AoC\_02 :  $c$  fails, any downstream AoC entries that assumes a well-defined causal order must be re-audited, downgraded, or retired in the Hall of Records.

# Abstract

In standard physics,  $c$  is defined as the speed of light—a transmission limit. In the 7-Dimensional Universe (7dU) framework, this definition is reversed: light travels at  $c$  because  $c$  is the structural limit of sequence.

Following the emergence of stable geometry ( $\pi$ ), the universe faces a secondary collapse risk: Simultaneity Failure. Without a fixed propagation slope, influences could traverse space instantaneously or variably, rendering causal order frame-dependent. To prevent this, the triadic interaction of Constraint ( $\zeta$ ) and Expansion ( $\omega$ ) locks into a rigid ratio, creating the Null Cone.

This paper frames  $c$  not as a velocity, but as the Causal Rigidifier. It is the invariant slope that prevents the past from overwriting the future. It allows  $\xi$  (Chance) to fluctuate without breaking the narrative of existence.

# 1. Collapse Consequence (Load-Bearing Statement)

This entry exists in the Atlas because without a finite, invariant null slope, ordered sequence cannot survive across geometry. If influences can propagate with zero cost, infinite speed, or frame-dependent limits, then before and after lose any stable meaning. In that regime, the rest of the Atlas is misnamed: constants cannot be survivors, because there is no coherent history in which to survive.

We therefore treat the existence of some finite, invariant null slope as a structural necessity once:

- closure of geometry has been established (AoC\_1\_On\_Pi), and
- a viability operator  $\hat{\Omega}$  has been fixed (AoC\_00).

Status: DERIVED: structural necessity of a finite, invariant null slope, given AoC\_00 (Collapse Operator) and AoC\_01 (On Pi). Numerical value and unit choice explicitly not claimed here.

## 1.1 Failure modes if no invariant null slope exists

DERIVED given AoC\_00, AoC\_01

We examine three generic failure modes for sequence when no finite, invariant null slope is selected.

### (a) Global simultaneity — instantaneous reach

If propagation is effectively instantaneous, any event can influence any other without delay. The universe admits only a smeared “Now”:

- every point in space is, in principle, causally connected to every other at zero cost;
- any attempted ordering of events can be scrambled by arbitrarily small perturbations.

For example, under small metric noise  $\xi \sim \delta g_{\mu\nu}$  or tiny variations in how observers slice spacetime into “instants,” events that are simultaneous in one admissible frame  $F$  can become non-simultaneous in another frame  $F'$ , allowing cycles where one observer can arrange  $A < B$  while another arranges  $B < A$ .

Under  $\hat{\Omega}$ , this regime fails viability: there is no robust way to maintain a partial order  $<$  that survives noise, because all events tend toward global simultaneity.

## (b) Frozen isolation — zero reach

If propagation speed is effectively zero, no event can influence any other. The universe fragments into isolated points:

- no interaction, no transport, no exchange of information;
- geometry may exist, but nothing happens within it.

Under  $\hat{\Omega}$ , this regime also fails viability: without any reach, there is no nontrivial sequence to preserve. All structures are trivial and dynamically dead.

## (c) Frame-dependent slope — order inversion

If the effective propagation slope depends on frame, location, or energy, different observers can disagree on the order of the same pair of events:

- in one admissible frame, event A precedes B;
- in another admissible frame, B precedes A.

This opens the door to causal loops: configurations where  $A < B < A$  becomes possible once frame changes and perturbations are combined. Under  $\hat{\Omega}$ , such loops are classified as identity failures: persistent structure cannot retain a consistent history.

## 1.2 Structural consequence

Across all three failure modes, the common result is:

- No stable partial order  $<$  on events survives  $\hat{\Omega}$ 's viability test.

Without a stable, noise-resistant  $<$ :

- atoms cannot have a well-defined sequence of binding interactions,
- stars cannot have a coherent formation and evolution history,
- observers cannot trace or record a consistent chain of experience.

Durable physical structure becomes a misnomer; there is only either static isolation, undifferentiated simultaneity, or looped self-contradiction. Thus, some finite, invariant null slope is required for the very notion of “survivor” in the Atlas to make sense.

Status: [DERIVED: structural consequence of failing to select a finite, invariant null slope. No claim is made here about the numerical value of (  $c$  ) or its unit realization.]

## 1a.1 Operator face

For this Atlas entry we use the recursive / viability face of the collapse operator defined in AoC\_00. In this face, the operator tests whether a candidate causal geometry can survive iterative stress (frame changes, metric noise) without losing a stable event ordering.

We specialize the generic form to:

$$\hat{\Omega}(M_n) := R(G(M_n), \xi)$$

where (see §§1a.3–1a.6 for full definitions):

- $M_n$  (defined in §1a.3) is the candidate state of the causal geometry at recursion depth  $n$ ;
- $G(M_n)$  (defined in §1a.4) is the geometry extractor that isolates the null-structure field and produces the dimensionless null-slope data used in this entry;
- $\xi$  (defined in §1a.5) is the noise / perturbation field (metric and frame jitter) that injects entropy into the geometry;
- $R(\cdot, \xi)$  (defined in §1a.6) is the viability map, returning one of:
  - Persist — the null-slope class converges to a finite, non-zero, frame-invariant fixed point and the induced event order remains acyclic;
  - Collapse — the structure freezes (no reach, effective isolation) or shatters (no coherent ordering possible);
  - Diverge — influence becomes unbounded or generates causal loops (order inversion under admissible perturbations).

In AoC\_2, a null-slope structure is a candidate survivor only if it is classified as Persist by this face of  $\hat{\Omega}$ .



Status: [DEFINITION: recursive / viability face of ( $\hat{\Omega}$ ) as instantiated for AoC\_02.]

## 1a.2 Domain — ANSATZ\_L (Lorentzian)

For AoC\_2 we work on a 4-dimensional differentiable manifold  $\mathcal{M}$  with the following structural properties. Together these define Ansatz\_L (the Lorentzian Ansatz) — they are inputs to this entry, not results of it.

- Signature (structure class).

$\mathcal{M}$  admits a Lorentzian metric of signature  $(-+++)$ : one timelike and three spacelike directions.

The existence of such a signature is assumed here; its necessity may be addressed in future Atlas entries (e.g. Selection Law). AoC\_02\_On\_c is conditional on this choice.

- Topology (causal well-posedness).

$\mathcal{M}$  is time-orientable (a consistent “future” direction can be defined globally) and globally hyperbolic (i.e., it admits a foliation by spacelike hypersurfaces supporting a well-posed initial value problem). This ensures that “histories” are meaningful objects to discuss.

- Metric freedom (no pre-picked geometry).

No specific metric tensor  $g_{\mu\nu}$  is fixed a priori beyond belonging to this Lorentzian class. The manifold is treated as a pre-metric stage that can support many candidate Lorentzian geometries; the Collapse Operator  $\hat{\Omega}$  acts to test which causal structures (if any) survive as viable.

Within AoC\_2, any claim about the necessity of a finite, invariant null slope  $k_*$  is therefore understood relative to this domain choice. If Ansatz\_L is later revised or falsified, this entry, AoC\_2..., must be re-audited under the updated domain.

Status: ANSATZ\_L: declared geometric domain

### 1a.3 State $M_n$ — DEFINITION

At recursion depth  $n$ , the system state is a candidate causal geometry encoded as the pair

$$M_n = (g_{\mu\nu}^{(n)}, \mathcal{C}^{(n)}).$$

Where:

- $g_{\mu\nu}^{(n)}$  (Metric Candidate).

A Lorentzian metric tensor field on  $\mathcal{M}$  at step  $n$ . It specifies distances, angles, and curvature, and belongs to the class of metrics compatible with Ansatz\_L in §1a.2.

- $\mathcal{C}^{(n)}$  (Causal Structure).

The field of future-directed light cones determined by  $g_{\mu\nu}^{(n)}$ . At each point

$p \in \mathcal{M}$ ,  $\mathcal{C}^{(n)}(p)$  is the set of all future-directed null directions through  $p$ , encoding which events can, in principle, influence which others. This induces a partial order  $\prec^{(n)}$  on events (“can causally reach”).

Compatibility Condition:

The metric and causal structure are required to be strictly compatible:

- A vector  $k^\mu$  at  $p$  lies on the light cone if  $g_{\mu\nu}^{(n)}(p) k^\mu k^\nu = 0$ ,  $k^\mu$  future-directed.
- The causal structure  $\mathcal{C}^{(n)}$  is nothing more and nothing less than the collection of these null directions across  $\mathcal{M}$ .

In the recursive application of the Collapse Operator  $\hat{\Omega}$ , we update

$$M_n = (g_{\mu\nu}^{(n)}, \mathcal{C}^{(n)}) \quad \longmapsto \quad M_{n+1} = \hat{\Omega}(M_n)$$

and test whether this sequence converges to a viable fixed point  $M_*$  whose null structure yields a finite, non-zero, frame-invariant null slope  $k_*$  and a stable, acyclic causal order  $\prec_*$ .

Status: DEFINITION: candidate state for  $\hat{\Omega}$ 's viability test

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## 1a.4 Geometry Extractor $G$ — DEFINITION

The geometry extractor  $G$  maps the full candidate state  $M_n$  to the specific structural data needed to test causal viability. It discards everything except the null structure and the associated null-slope behaviour.

### 1. Extracting the Null Structure

Given the metric candidate  $g_{\mu\nu}^{(n)}$ ,  $G$  first identifies, at each point  $p \in \mathcal{M}$ , the set of all future-directed null vectors:

$$\mathcal{N}^{(n)}(p) = \left\{ k^\mu \neq 0 \mid g_{\mu\nu}^{(n)}(p) k^\mu k^\nu = 0, k^\mu \text{ future-directed} \right\}.$$

The collection  $\mathcal{N}^{(n)} = \{ \mathcal{N}^{(n)}(p) \}_{p \in \mathcal{M}}$  is the null structure field. It encodes the local “influence boundary” without committing to any unit system.

### 2. Coordinate Slopes from Null Directions

To probe causal sequencing,  $G$  examines how  $\mathcal{N}^{(n)}$  appears in admissible local charts.

In a local coordinate chart  $\chi = (t, x^1, x^2, x^3)$  adapted to the time orientation, each null vector  $k^\mu \in \mathcal{N}^{(n)}(p)$  determines a coordinate slope

$$v^{(n)}(p, \chi; k) = \frac{dx^i}{dt},$$

where  $x^i$  is the spatial coordinate direction aligned with the spatial projection of  $k^\mu$  in the chart  $\chi$ .

These slopes are chart-dependent; they represent the “speed of light” as seen in that particular frame, not yet a survivor.

Define

$\mathcal{S}^{(n)}(p, \chi) = \{v^{(n)}(p, \chi; k) \mid k \in \mathcal{N}^{(n)}(p)\}$  as the set of such null slopes in chart  $\chi$  at point  $p$ .

### 3. Null-Slope Class and Survivor Candidate $k_*$

The role of  $G$  is not to assign a numerical value to  $c$ , but to provide the null-slope class that the viability map  $R$  will test.

We define the null-slope class at depth  $n$  as

$$\mathcal{K}^{(n)} = \left\{ \mathcal{S}^{(n)}(p, \chi) \mid p \in \mathcal{M}, \chi \text{ admissible local chart} \right\}.$$

From this class,  $R$  will ask whether there exists a survivor pattern  $k_*$  such that:

- Finite: all null slopes are bounded in magnitude in every admissible chart,

$$0 < \left| v^{(n)}(p, \chi; k) \right| < \infty \quad \forall p, \chi, k.$$

- Non-zero: propagation is not frozen ( $v^{(n)} \neq 0$ ).
- Frame-stable: the relation between causal ordering and these null slopes is preserved across all admissible frames (no choice of chart converts an acyclic order into a causal loop).

Operationally,  $G(M_n)$  is the data package

$$G(M_n) = (\mathcal{N}^{(n)}, \mathcal{K}^{(n)}),$$

which supplies the null-cone geometry and its induced slope behaviour to the viability map  $R(\cdot, \xi)$  in §1a.6.

Status: [DEFINITION: extraction of null-structure and null-slope class for  $\hat{\Omega}$ 's viability test]

## 1a.5 Noise / Perturbation $\xi$ — [ANSATZ\_ $\xi$ ]

The entropy-injection field  $\xi$  is the operational face of Chance ( $\xi$ ) in this entry. It supplies the structural stress that distinguishes a fragile coincidence from a robust Survivor. A candidate null-slope structure counts as a Survivor only if it remains viable under this class of perturbations.

We model  $\xi$  as a parametrized family of perturbations acting on the state  $M_n$ :

$$\xi(\lambda) : M_n \mapsto M_n(\lambda),$$

where  $\lambda$  encodes the noise parameters (amplitude, bandwidth, and, in principle, random seed). The perturbation consists of two coupled components:

### 1. Metric Fluctuations (Geometry Stress)

To test robustness against local geometric deformation,  $\xi$  injects small stochastic perturbations into the metric tensor:

$$g_{\mu\nu}^{(n)} \longrightarrow g_{\mu\nu}^{(n)}(\lambda) = g_{\mu\nu}^{(n)} + \varepsilon_\xi h_{\mu\nu}(\lambda),$$

where:

- $\varepsilon_\xi > 0$  is the noise amplitude (assumed small and bounded),
- $h_{\mu\nu}(\lambda)$  is a random symmetric tensor field drawn from a bounded distribution and smooth enough to preserve differentiability.

Purpose. These fluctuations “fuzz” the light cones and test whether the partial order  $<$  induced by the null structure remains acyclic. If arbitrarily small metric perturbations can produce loops (e.g.  $A < B$  and  $B < A$  in some perturbed realization), the candidate null-slope structure fails viability.

## 2. Frame Jitter (Invariance Stress)

To test whether the null-slope behaviour is truly physical and not a coordinate artefact,  $\xi$  also applies random, admissible changes of chart that play the role of local “boost-like” transformations.

Formally, for each  $\lambda$  we consider a family of diffeomorphisms

$$\Phi_\lambda : \mathcal{M} \rightarrow \mathcal{M},$$

compatible with Ansatz\_L (they preserve the Lorentzian signature and time orientation), and examine the transformed description of the same perturbed state:

$$M_n(\lambda) \mapsto \Phi_\lambda^* M_n(\lambda).$$

Purpose. This component tests frame-stability: if the null-slope class  $\mathcal{K}^{(n)}$  appears well-behaved in one admissible chart but, under allowed  $\Phi_\lambda$ , produces order inversions or causal loops, then the structure is not a Survivor. A viable  $k_*$  must support the same acyclic  $<$  in all admissible frames.

## 3. Stress Regime and Role in the Recursion

At each recursion step,  $\xi$  supplies a controlled but nontrivial stress regime:

- Amplitude:  $\varepsilon_\xi$  is chosen small enough to remain in the same qualitative geometric regime, but large enough that fine-tuned coincidences are unlikely to persist.
- Bandwidth: the spectrum of perturbations (in scale and “boost-like” parameters) is broad enough to probe local and mesoscale structure, not just a single mode.

The recursive question posed by the operator is: Does the null-slope structure encoded in  $G(M_n)$  persist as a fixed-point pattern under all  $\xi(\lambda)$  in the declared class?

If it does,  $M_n$  is eligible for the Persist label in §1a.6. If small, admissible perturbations routinely generate causal loops or destroy the stability of the null-slope class, the candidate is classified as Collapse or Diverge.

Status: [ANSATZ\_ $\xi$ : Declared class of structural stress (metric fluctuations + frame jitter)]

## 2. Scope and Non-Claims

This section delimits what AoC\_02\_On\_c is actually responsible for. It separates the structural claim (a necessary invariant) from anything that lives in metrology or phenomenology.

### 2.1 This entry does claim — SCOPE

Within the framework of the Prooffield Index and given its declared preconditions, this entry does claim the following:

#### 1. Existence of a structural null-slope invariant

There exists a finite, non-zero, dimensionless null-slope class  $[k_*]$  such that the Collapse Operator  $\hat{\Omega}$  (in its recursive / viability face, §1a) selects it as a Persist fixed point (§1a.6) for any universe that:

- Has spatial closure — i.e. satisfies the geometric stability established in AoC\_01\_On\_Pi ( $\pi$ ).
- Inhabits a Lorentzian domain — i.e. lives on a manifold satisfying Ansatz\_L (§1a.2: signature  $(-+++)$ , time-orientable, globally hyperbolic).
- Supports phenomenological causality — i.e. satisfies the Phenomenological Sequence Requirement: observers can stably order events into histories (PHENOMENOLOGICAL REQUIREMENT).

Under these conditions,  $\hat{\Omega}$  does not admit  $k \rightarrow 0$ ,  $k \rightarrow \infty$ , or frame-dependent  $k$  as survivors. The only non-trivial survivors are fixed-point null-slope classes  $[k_*]$  that remain finite, non-zero, and frame-stable under the  $\xi$ -stress defined in §1a.5.

#### 2. Structural necessity (conditional)

For universes satisfying  $\{\text{AoC\_00, AoC\_01, Ansatz\_L, Phenomenological Sequence Requirement, Ansatz\_}\xi\}$ , the existence of such a  $[k_*]$  is necessary for durable histories. Any candidate geometry in this domain either:

- acquires a finite, invariant null-slope class  $[k_*]$  and is labeled Persist, or
- falls into Collapse or Diverge as defined in §1a.6 (frozen isolation, orderless smear, runaway slopes, or causal loops).

All necessity claims in this entry are explicitly conditional on these preconditions. If any of AoC\_01 (On Pi), Ansatz\_L, the Phenomenological Sequence Requirement, or Ansatz\_ξ is later revised or falsified, AoC\_2 (On c) must be re-audited.

## 2.2 This entry explicitly does not claim — NON-CLAIMS

To comply with the Units Firewall (AoC\_00 §5) and avoid category errors, this entry does not claim:

### 1. No metrological value

We do not derive the numerical value:

$$c \approx 299\,792\,458 \text{ m/s.}$$

That number is a consequence of how humans chose to define the meter and the second. AoC\_02 derives a dimensionless survivor  $[k_*]$ , not a calibrated SI speed.

### 2. No unit or standard redefinition

We do not propose any revision of SI units, metrological conventions, or measurement protocols. Whatever value metrologists assign to  $c$  in their chosen units lies outside the jurisdiction of this entry.

### 3. No exotic vacuum claims

We do not:

- assert the existence of superluminal signaling in vacuum,
- derive energy-dependent vacuum dispersion, or
- introduce new propagation modes beyond the standard null-cone structure.

AoC\_2 is about light-cone rigidity, not about adding new particles or channels.

### 4. No signature derivation

We do not derive the Lorentzian signature  $(-+++)$ .



It is taken as Ansatz\_L (§1a.2): a declared domain choice for this entry. The question of why this signature may be preferred or selected over alternatives is reserved for other Atlas work (e.g. Selection Law / dimensional studies), not for AoC\_2.

Tag: Structure, not metrology.

AoC\_02 derives the necessity of a finite, invariant null slope  $[k_*]$  under  $\hat{\Omega}$ , not the human label “299 792 458 m/s.”

### 3. Preconditions / Ansätze

This section is the dependency manifest for AoC\_2.

Every load-bearing premise is named, tagged, and scoped so there are no hidden axioms in the derivation.

If any of these premises fail, the conclusions of this entry do not hold.

#### 3.1 AoC Dependencies

These are structural foundations established elsewhere in the Atlas.

##### 1. AoC\_00 — The Collapse Operator $\hat{\Omega}$ — DERIVED

We assume:

- The validity of the Collapse Operator  $\hat{\Omega}$  as defined in AoC\_00.
- The recursive / viability face used here (§1a.1).
- The Persist / Collapse / Diverge classification (§1a.6).
- The Units Firewall discipline (dimensionless survivors, no unit leakage).

All uses of  $\hat{\Omega}$ ,  $\xi$ , and the viability map  $R(\cdot)$  in this entry depend on the definitions and proofs given in AoC\_00.

##### 2. AoC\_01 — On $\pi$ (Spatial Closure) — DERIVED

We assume that:

- Stable geometric closure exists in 2D/3D.
- The manifold can support coherent spatial loops and curvature.

Without a survivable spatial manifold to propagate through, the notion of a propagation slope (and thus “speed”) is undefined. AoC\_02 is downstream of this closure.

## 3.2 Additional Ansätze

These are entry-specific assumptions. They are inputs, not conclusions.

### 1. Ansatz\_L — Lorentzian Signature — ANSATZ

We assume the spacetime domain of interest:

- Admits a Lorentzian metric signature  $(-+++)$ : one timelike, three spacelike directions.
- Is time-orientable (a consistent “future” direction exists globally).
- Is globally hyperbolic (admits spacelike Cauchy slices and well-posed initial value problems).

Scope note: Why the universe is Lorentzian rather than, say, Euclidean  $(++++)$  or ultra-hyperbolic is delegated to selection and dimensionality work (e.g. Extension Bounds / Selection Law). For AoC\_02, Ansatz\_L is a declared constraint.

### 2. Ansatz\_B — Boost / Frame Admissibility — ANSATZ

We assume:

- There exists a non-trivial family of admissible frame transformations (“boost-like” maps) that send valid observers to valid observers.
- Physics is not confined to a single privileged frame.

Implication: Any candidate survivor that is “stable” only in one preferred frame, but produces causal loops or pathologies when viewed from other admissible frames, fails  $\hat{\Omega}$ ’s viability test. Frame-robustness is part of survivorship.

### 3. Phenomenological Sequence Requirement — PHENOMENOLOGICAL REQUIREMENT

We take as observational input that:

- In the universe we inhabit, events can be stably ordered into histories.
- “Before” and “after” are physically distinguishable for observers; causality is a lived phenomenon, not merely a mathematical option.

Constraint: Any candidate geometry that yields:

- global simultaneity (no meaningful sequence),
  - frozen isolation (no change), or
  - generic causal loops ( $A < B < A$  for ordinary observers), is treated as physically inadmissible for our domain. AoC\_02 is scoped to universes with lived, stable causality.
4.     Ansatz\_ $\xi$  — Noise Class — ANSATZ

We assume:

- Metric fluctuations and frame jitter are small but non-zero, as defined in §1a.5.
- $\xi$  explores a bounded but non-trivial neighborhood of each candidate state  $M_n$ .

Implication: Fine-tuned geometries that survive only at exactly  $\xi = 0$  are rejected. A Survivor must withstand the “shaking” induced by  $\xi$  without losing its null-slope structure or acyclic order.

### 3.3 Failure of Preconditions — JURISDICTION LIMIT

The necessity claims of AoC\_02 hold only within the intersection of the above conditions.

- If AoC\_00 or AoC\_01 are falsified, the structural backbone of the derivation fails, and this entry loses its foundation.
- If Ansatz\_L is false (e.g. the underlying universe is fundamentally Euclidean or has a different signature), AoC\_02 is simply out of jurisdiction.
- If the Phenomenological Sequence Requirement does not apply (e.g. in a universe where time is not a meaningful or experienced order), then the demand for a survivable sequential structure is void.

In all such cases, AoC\_02 should be treated not as “wrong,” but as inapplicable outside its declared domain.

## 4. Structural Problem: Simultaneity vs Sequence

### 4.1 Problem Statement — DERIVED given §3

With AoC\_01 ( $\pi$ ) in place, the Atlas has Spatial Closure: a geometric manifold capable of sustaining stable, closed loops (curvature). We know where structures can exist.

However, spatial closure alone produces a static block. To satisfy the Phenomenological Sequence Requirement (§3.2.3), the universe must also support evolution: a stable, directed ordering of events into histories.

The key structural point is:

Sequence is not guaranteed by geometry alone.

Even on a Lorentzian manifold (Ansatz\_L), causal order is underdetermined unless the propagation structure is constrained. In particular, the behavior of the null-slope class  $K^{(n)}$  (defined in §1a.4) determines whether a stable partial order  $<$  survives the recursive viability test  $\hat{\Omega}$  (§1a).

If propagation is:

- too unconstrained (effective null slope  $k \rightarrow \infty$ ): Extension ( $\omega$ ) dominates and influence is effectively instantaneous,
- too constrained (effective null slope  $k \rightarrow 0$ ): Constraint ( $\zeta$ ) dominates and influence cannot propagate, or
- incoherently variable (null slope  $k$  depends on frame, position, or energy),

then the partial order  $<$  either fails to exist or fails to remain acyclic under admissible transformations and  $\xi$ -perturbations.

In such cases,  $\hat{\Omega}$  must classify the candidate geometry as Collapse or Diverge (§1a.6), and no Survivor exists.

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## 4.2 Pathologies Without an Invariant Null Slope — DERIVED given §1, §1a, §3

We now examine the behavior of the null-slope class  $K^{(n)}$  when there is no finite, invariant fixed point  $k_*$ . The three broad failure regimes correspond to the limiting or drifting behavior of  $k$ .

### Case A: Unconstrained Extension — $k \rightarrow \infty$ (Simultaneity Failure)

- Effective Regime: Influence can traverse arbitrary spatial separations in arbitrarily small coordinate time (instantaneous reach in some frame).
- Geometric Picture: In a given chart, the null cones open toward spacelike directions so strongly that the effective “speed limit” becomes arbitrarily large.
- Structural Failure:
  - In one admissible frame  $\mathcal{F}$ , a set of events may be treated as effectively simultaneous, because signals can connect them in negligible coordinate time.
  - Under an admissible frame transformation (Ansatz\_B) combined with  $\xi$ -induced metric/frame jitter (§1a.5), the same events are seen in a different frame  $\mathcal{F}'$  with a tilted notion of simultaneity.
  - A sequence that was “all at once” in  $\mathcal{F}$  can become partially ordered in  $\mathcal{F}'$ , and small perturbations can swap the apparent order of events.
- $\hat{\Omega}$  Verdict:
  - Under iteration, the partial order  $\prec$  is not stable: arbitrarily small  $\xi$ -perturbations can change or invert perceived order.
  - The family  $K^{(n)}$  has no finite fixed point; it effectively drifts toward unbounded slope.
  - Classification: Diverge (Simultaneity Failure). No robust, observer-independent history survives.

## Case B: Over-Constraint / Isolation — $k \rightarrow 0$ (Interaction Failure)

- Effective Regime: Propagation speed tends toward zero; influence cannot traverse spatial intervals.
- Geometric Picture: In each chart, null cones close onto the time-like axis: signals can move “forward in time” but not outward in space.
- Structural Failure:
  - Each world-line becomes effectively causally isolated from its neighbors.
  - No event can influence any spatially separated event; there are no non-trivial causal chains connecting distinct regions.
  - The universe degenerates into a dust of non-communicating points; curvature and structure may exist mathematically, but no dynamics can couple them.
- $\hat{\Omega}$  Verdict:
  - Under  $\xi$ -perturbations,  $K^{(n)}$  still collapses toward zero reach; no non-trivial partial order  $<$  connecting distinct events can be maintained.
  - Classification: Collapse (Interaction Failure). History cannot form because nothing can happen between distinct locations.

## Case C: Variable / Drifting Null Slope — $k = k(x, E, \text{frame})$ (Identity Failure)

- Effective Regime: The effective maximum propagation slope depends on:
  - the observer’s frame (boost),
  - the spacetime location, and/or
  - properties of the probe (e.g. energy).
- Geometric Picture:
  - Different frames or regions see different light-cone openings, so the boundary between “can influence” and “cannot influence” is not universal.

- Structural Failure:
  - If  $k$  depends on frame:
  - In frame  $\mathcal{F}$ , events satisfy  $A < B$  (A can influence B).
  - In another admissible frame  $\mathcal{F}'$ , with a different effective  $k$ , the same pair may satisfy  $B < A$  or become incomparable.
  - Under  $\xi$ -driven frame jitter (§1a.5), loops of the form  $A < B < A$  can appear.
  - If  $k$  depends strongly on position/energy:
  - The partial order  $<$  becomes patch-dependent; chains that appear consistent locally cannot be extended globally without contradiction.
- $\hat{\Omega}$  Verdict:
  - The recursive application of  $\hat{\Omega}$  detects causal loops or non-convergent behavior in  $<$ .
  - The family  $K^{(n)}$  fails to converge to a single, frame-stable class.
  - Classification: Diverge (Identity Failure). The identity of “cause precedes effect” is not preserved.

### 4.3 Target of AoC\_2 — DERIVED given §1, §1a, §3

Within the declared domain (§3), these three regimes exhaust the ways in which the null-slope class  $K^{(n)}$  can fail to support survivable sequence:

- Unbounded slope:  $k \rightarrow \infty \rightarrow$  Simultaneity Failure (Case A).
- Vanishing slope:  $k \rightarrow 0 \rightarrow$  Interaction Failure (Case B).
- Drifting / non-invariant slope:  $k = k(x, E, \text{frame}) \rightarrow$  Identity Failure (Case C).

The goal of AoC\_02\_On\_c is to show that, given:

- AoC\_00 and AoC\_01,
- Ansatz\_L (Lorentzian domain),
- Ansatz\_B (admissible frames),
- Ansatz\_ξ (non-zero noise), and
- the Phenomenological Sequence Requirement,

the Collapse Operator  $\hat{\Omega}$  inevitably filters out all three pathological regimes, and that the only states labeled Persist by  $R(\cdot)$  (§1a.6) are those for which:

- $k_*$  is finite (avoiding Simultaneity Failure),
- $k_* \neq 0$  (avoiding Interaction Failure), and
- $k_*$  is frame-invariant (avoiding Identity Failure).

This finite, non-zero, frame-invariant fixed point  $k_*$  is the Survivor null-slope structure.

In the Atlas, once calibrated to human units, it is what we call the speed of light  $c$ .



## 5. Derivation Program (Recursive Face)

Status: PROGRAM given §1a, §3, §4

The goal of this section is to apply the recursive / viability face of the Collapse Operator,  $\hat{\Omega}$ , (defined in §1a.1) to the candidate causal state  $M_n$ , and to show that under the declared preconditions (§3):

- all geometries exhibiting the pathologies of §4 (Cases A, B, C) are driven to Collapse or Diverge, and
- the only geometries that return Persist under the viability map  $R$  (§1a.6) are those admitting a finite, non-zero, frame-invariant null-slope fixed point  $k_*$ .

We do not attempt to solve a closed-form operator equation for  $\hat{\Omega}$  (which would require explicit functional forms for the metric dynamics and noise field). Instead, we:

1. Define what it means for a causal sequence to be survivable under  $\hat{\Omega}$ .
2. Start from a generic candidate state  $M_0$  and its null-slope class  $K^{(0)}$ .
3. Apply the recursion  $M_{n+1} = \hat{\Omega}(M_n)$  with noise  $\xi$  and admissible frame transformations (Ansatz\_B).
4. Track the fate of  $K^{(n)}$  under this recursion, showing that:
  - $k \rightarrow 0, k \rightarrow \infty$ , or non-invariant  $k$  all violate survivability, and
  - only a finite, non-zero, frame-invariant fixed point  $k_*$  can satisfy the survivability criteria.

In other words, this section is the logical flow from “generic Lorentzian candidate” to “unique, survivable null-slope structure,” using the previously defined Viability Map  $R$  (§1a.6) as the decision engine.

---

## 5.1 Survivable Sequence — DEFINITION

We now formalize what it means for a causal ordering of events to survive under the action of  $\hat{\Omega}$ , given the domain and Ansätze in §3.

Let  $\mathcal{E}$  be the set of spacetime events, and let  $\prec$  be a candidate causal relation on  $\mathcal{E}$ .

We say that  $\prec$  is a Survivable Sequence if and only if all three of the following conditions hold:

### 1. Frame-Local Consistency (Acyclicity)

For every admissible frame  $\mathcal{F}$  (coordinate chart  $\chi$ ):

- The projection of  $\prec$  onto  $\mathcal{F}$  is a strict partial order:
- Transitive: If  $A \prec B$  and  $B \prec C$ , then  $A \prec C$ .
- Irreflexive (acyclic): There is no event  $A$  such that  $A \prec A$  (no closed causal loops within that frame).

This ensures that, in any single frame, histories can be written as non-circular chains of events.

### 2. Transformation Robustness (Ansatz\_B)

The relation  $\prec$  is invariant under the admissible family of frame transformations (boost-like maps) declared in Ansatz\_B (§3.2.2):

- If  $A \prec B$  in frame  $\mathcal{F}$ , then for any admissible transformation to frame  $\mathcal{F}'$ , the transformed events still satisfy  $A \prec B$  (or at worst remain incomparable; they must never satisfy  $B \prec A$ ).

Operational implication:

No allowed change of frame may invert a causal ordering. What is “before” for one admissible observer cannot become “after” for another.

### 3. Perturbation Stability (Ansatz $\xi$ )

The relation  $<$  remains consistent under the class of perturbations  $\xi$  defined in §1a.5 (metric fluctuations + frame jitter):

- If  $A < B$  in the unperturbed state  $M_n$ , then for any perturbed state  $M_n(\lambda)$  within the declared noise bandwidth  $(\epsilon_\xi, \lambda)$ , it must remain true that:
  - either  $A < B$  still holds, or
  - $A$  and  $B$  become incomparable, but never  $B < A$ .

#### Operational implication:

The causal sequence is not a fine-tuned artifact that disappears or reverses under arbitrarily small  $\xi$ -perturbations. It is a robust feature of the geometry, not a numerical accident at  $\xi = 0$ .

---

When all three conditions are satisfied, we say that  $<$  survives the action of  $\hat{\Omega}$  and is eligible for a Persist classification by the viability map  $R$  (§1a.6). Subsequent subsections (§5.2–§5.4) will apply these criteria to the families of null-slope behaviors identified in §4, and show that only a finite, non-zero, frame-invariant fixed point  $k_*$  can support a Survivable Sequence.5.2 Start from a Generic Candidate — SETUP

To show that the invariant null slope is a necessary survivor rather than a hidden assumption, we must begin from a generic state in the declared domain and let the Collapse Operator  $\hat{\Omega}$  do the selection.

We therefore choose an arbitrary initial state

$$M_0 = (g_{\mu\nu}^{(0)}, C^{(0)})$$

subject only to the constraints of Ansatz\_L (§1a.2).

---

## 5.2.1 Properties of the Generic Initial State

At recursion depth  $n = 0$ , we assume only the minimal conditions required by the Lorentzian domain:

- Signature:

$g_{\mu\nu}^{(0)}$  is a Lorentzian metric of signature  $(-, +, +, +)$  everywhere.

- Regularity:

$g_{\mu\nu}^{(0)}$  is smooth enough to define light cones and null tangents at every point.

- No Pre-tuning of the Null Slope:

We do not assume that the null cones are:

- rigid,
- uniform across the manifold, or
- frame-invariant under admissible boosts.

In particular, in this generic state:

- the effective propagation “speed” may depend on position  $x$ ,
- it may depend on the observer’s frame (boost parameter),
- it may depend on energy or other local fields,
- and local cones may be very wide (near  $k \rightarrow \infty$ ) or very narrow near ( $k \rightarrow 0$ ).

In other words, the initial geometry is allowed to explore all three pathological regimes from §4:

- Simultaneity-like: near-instant propagation (Case A),
- Isolation-like: near-frozen propagation (Case B),
- Drift-like: frame- or position-dependent propagation (Case C),

as well as any mixture of these.

---

## 5.2.2 Initial Geometry Extraction

We apply the geometry extractor  $G$  from §1a.4 to obtain the initial null structure and slope data:

$$G(M_0) = (N^{(0)}, K^{(0)}).$$

Here:

- $N^{(0)}$  is the field of null directions (the candidate light cones), and
- $K^{(0)}$  is the null-slope class at  $n = 0$ , which may be a highly irregular set of slopes  $k^{(0)}$  varying across spacetime and between frames.

At this stage,  $K^{(0)}$  is intentionally “wild”: we place no restrictions on it beyond those already implied by §3 (AoC dependencies + Ansätze). We do not build in finiteness, non-zerosness, or invariance.

## 5.2.3 Launching the Recursion

From this generic starting point, we iterate the Collapse Operator:

$$M_{n+1} = \hat{\Omega}(M_n), \quad n = 0, 1, 2, \dots$$

At each step:

1. Apply the noise class  $\xi$  (metric fluctuations + frame jitter, §1a.5) to generate perturbed states  $M_n(\lambda)$ .
2. Extract geometry via  $G$ :

$$G(M_n(\lambda)) = (N^{(n)}(\lambda), K^{(n)}(\lambda)).$$

3. Evaluate viability using the map  $R$  (§1a.6):

$$R(\{G(M_n(\lambda))\}_{\lambda \in \Lambda}) \in \{\text{Persist, Collapse, Diverge}\}.$$

The program of §§5.3–5.4 is to show:

- that all trajectories in which  $K^{(n)}$  flows toward:
  - $k \rightarrow \infty$  (Case A),
  - $k \rightarrow 0$  (Case B),
  - or frame-/position-dependent  $k$  (Case C),  
are classified by R as Collapse or Diverge, and
- that the only trajectories which can ever be labeled Persist are those in which  $K^{(n)}$  converges to a finite, non-zero, frame-invariant fixed class  $k_*$ .

Status: SETUP (Initialization of the recursive proof; no assumptions yet made about the form or value of  $k_*$ )

## 5.3 Run Through Unstable Regimes — [EXCLUSION]

We now trace the fate of the candidate null-slope class  $K^{(n)}$  when the recursion drives it toward any of the pathological regimes identified in §4. For each case, we apply the Survivable Sequence criteria (§5.1) and the Viability Map R (§1a.6) to decide between Persist, Collapse, and Diverge.

### 5.3.1 Rejection of Unbounded Extension (Case A: $k \rightarrow \infty$ )

Hypothesis.

Suppose the recursive evolution of  $M_n$  drives  $K^{(n)}$  toward unbounded extension, i.e. an effective propagation speed that becomes arbitrarily large (the Newtonian / Galilean “instantaneous influence” limit).

#### 1. Mechanism of Failure (Simultaneity Tipping).

In some admissible frame  $F$ , let events  $A$  and  $B$  be spatially separated and joined by an “instantaneous” signal, so that in  $F$  they are effectively simultaneous:

$$A \sim B \quad \text{with} \quad v^{(n)}(F; A \rightarrow B) \approx \infty.$$

Under the noise field  $\xi$  (§1a.5), which includes small, random boost-like transformations, the hypersurface of simultaneity tilts. In a perturbed frame  $F'$ , the same pair can acquire a time ordering:

$$A < B \text{ in } F, \not\equiv B < A \text{ in } F'.$$

## 2. Violation of Survivability.

This inversion violates:

- Condition 2 (Transformation Robustness, §5.1): the order of  $A$  and  $B$  is not invariant under admissible frame transformations; and
- Condition 3 (Perturbation Stability, §5.1): an arbitrarily small  $\xi$ -perturbation can flip  $A < B$  into  $B < A$ .

## 3. Viability Verdict.

The partial order  $<$  fails to be invariant and acyclic under the combined action of  $\xi$  and admissible boosts. The Viability Map returns:

$$R(G(M_n), \xi) = \text{Diverge} \quad (\text{Simultaneity Failure}).$$

Case A is therefore excluded as a Survivable Sequence.

## 5.3.2 Rejection of Zero Extension (Case B: $k \rightarrow 0$ )

Hypothesis.

Suppose the recursive evolution of  $M_n$  drives  $K^{(n)}$  toward zero extension, i.e. the light cones collapse onto the timelike axis and effective propagation speed tends to zero.

### 1. Mechanism of Failure (Isolation).

In this regime, the causal future of any event  $A$  shrinks to its own worldline:

$$J^+(A) \approx \{\text{future points of } A \text{ on the same worldline}\}, \quad \Delta x \rightarrow 0 \text{ for all causal links.}$$

No influence can traverse finite spatial separations; the manifold decomposes into a dust of non-communicating worldlines.

## 2. Violation of Survivability.

- Condition 1 (Frame-Local Consistency, §5.1) is trivially satisfied (no loops; there are barely any arrows at all).
- But the resulting order is degenerate: there is no non-trivial partial order  $\prec$  connecting distinct spatial points.
- This contradicts the Phenomenological Sequence Requirement (§3.2.3): a universe with  $k \rightarrow 0$  cannot host the lived notion of history in which distinct events are causally linked. It also violates the implicit drive of the Extension operator  $\hat{\Omega}$ , which demands non-zero reach.

## 3. Viability Verdict.

Under  $\hat{\Omega}$ , such a geometry is classified as:

$$R(G(M_n), \xi) = \text{Collapse} \quad (\text{Interaction Failure}).$$

The system “freezes” into trivial, isolated threads rather than a coherent, evolving

history. Case B is therefore excluded as a Survivable Sequence in any universe satisfying §3.2.3.

### 5.3.3 Rejection of Variable / Drifting Slope (Case C: $k = k(F, x, E)$ )

Hypothesis.

Suppose the recursive evolution drives  $K^{(n)}$  toward a finite but non-invariant regime, where the effective null slope depends on frame, position, or energy:

$$k = k(F, x, E).$$

Examples include aether theories with a preferred frame, or vacua with strong dispersion.

#### 1. Mechanism of Failure (Ordering Paradox).

If two admissible frames  $F_1$  and  $F_2$  measure different effective speed limits, they define different null cone boundaries. For some event pair (A,B):

- In  $F_1$ : A lies inside the future cone of B, so  $B \prec A$ .



- In  $F_2$ : due to a different  $k$ , the same pair may be spacelike or even timelike-reversed, yielding  $A < B$  or no clear order.

By chaining such frame-dependent links (possibly in the presence of small  $\xi$ -perturbations), one can construct a closed causal loop:

$$A < B < C < A.$$

2. Violation of Survivability.

- This directly violates Condition 1 (Acyclicity, §5.1): a closed loop  $A < \dots < A$  appears.
- It also violates Condition 2 (Transformation Robustness, §5.1): the causal relation between events depends on which admissible frame is chosen.

3. Viability Verdict.

The partial order  $<$  becomes observer-dependent and cyclic. The Viability Map returns:

$$R(G(M_n), \xi) = \text{Diverge} \quad (\text{Identity Failure}).$$

Case C is therefore excluded as a Survivable Sequence.

Status:

[EXCLUSION: All trajectories with  $k \rightarrow 0$ ,  $k \rightarrow \infty$ , or frame/drift-dependent  $k$  are classified by  $R$  as Collapse or Diverge. Only fixed, finite, non-zero, frame-invariant candidates remain for §5.4.]

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## 6. Stress Tests & Survivability Window ( $\zeta$ , $\omega$ , $\xi$ )

To certify the survivor as a structural invariant rather than a local accident, we reconnect the AoC\_002 derivation to the canonical operator channels defined in AoC\_00 §2. We show that the invariant null slope  $k_*$  exists only within a bounded Survivability Window where Constraint ( $\zeta$ ), Extension ( $\omega$ ), and Chance ( $\xi$ ) are in balance.

### 6.1 Link to Operator Channels — [MAPPING]

We interpret the operator channels of  $\hat{\Omega}$  in the specific domain of causal geometry:

- $\zeta$  — Constraint  $\rightarrow$  Locality / Finite Speed Limit

The Constraint channel suppresses non-local influence. It imposes a “cost” for spatial reach per unit time.

- High  $\zeta$ : Strong locality; effective speed limit is low (cones narrow).
- Low  $\zeta$ : Weak locality; effective speed limit is high (cones widen).
- $\zeta \rightarrow 0$ : Instantaneous action-at-a-distance (flattened cones; Newtonian limit).

- $\omega$  — Extension  $\rightarrow$  Reach / Propagation Drive

The Extension channel encodes the drive to connect spatially separated events. It demands that influence can traverse the manifold.

- High  $\omega$ : Strong drive to connect distant regions.
- $\omega \rightarrow 0$ : No drive to connect; the universe tends toward frozen isolation (no propagation).

- $\xi$  — Chance  $\rightarrow$  Metric & Frame Jitter

As defined in §1a.5, this channel injects structured entropy (metric fluctuations + frame jitter) to test whether the causal order and null-slope class are robust or merely fine-tuned.

In this language, the invariant null slope  $k_*$  from §5 is the balanced outcome of the tension between  $\zeta$  (which limits speed),  $\omega$  (which demands reach), and  $\xi$  (which refuses fine-tuned, fragile solutions).

## 6.2 Perturbation Table — [DERIVED]

We now summarize the behavior found in §§5.3–5.4 in the canonical stress-test format of AoC\_00 §6.1, expressing each failure mode and the survivor in terms of  $\zeta$ ,  $\omega$ ,  $\xi$ .

| Parameter Limit                            | Effective Regime         | Fate of $k$                                                   | Fate of $<$                                                          | $\hat{\Omega}$ Classification  |
|--------------------------------------------|--------------------------|---------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------|
| $\zeta \rightarrow 0$ (Constraint fails)   | Unbounded Extension      | $k \rightarrow \infty$ (flattened cones; “instant” reach)     | Order flips under boosts + $\xi$ ; loops appear                      | Diverge (Simultaneity Failure) |
| $\omega \rightarrow 0$ (Extension fails)   | Zero Reach               | $k \rightarrow 0$ (no propagation across space)               | Only trivial self-order; no inter-point links                        | Collapse (Interaction Failure) |
| $\xi \rightarrow 0$ (Chance fails)         | Sterile Determinism      | Degenerate set of candidate $k$ ’s not separated by selection | Multiple structurally distinct orders compatible; no unique survivor | Diverge (No Selection)         |
| $\xi \gg \zeta, \omega$ (Chance dominates) | Noise-Dominated Geometry | $k$ fluctuates wildly; cones lose stable boundary             | Order becomes jittery / inconsistent; identity of $<$ not preserved  | Diverge (Identity Death)       |
| Balanced ( $\zeta, \omega, \xi$ )          | Survivable Window        | $k \rightarrow k_*$ (finite, non-zero, frame-invariant)       | \prec acyclic, frame-robust, $\xi$ -stable                           | Persist (Coherence)            |

In words:

- If Constraint vanishes ( $\zeta \rightarrow 0$ ), we recover Case A of §5.3: unbounded extension, flattened cones, and simultaneity failure  $\rightarrow$  Diverge.
- If Extension vanishes ( $\omega \rightarrow 0$ ), we recover Case B: zero reach, frozen isolation  $\rightarrow$  Collapse.
- If Chance vanishes ( $\xi \rightarrow 0$ ),  $\omega$  cannot discriminate between competing candidate geometries; degeneracy is never lifted  $\rightarrow$  Diverge (no selection).
- If Chance dominates ( $\xi \gg \zeta, \omega$ ), structure itself is shaken apart; cones and order lose identity  $\rightarrow$  Diverge (identity death).
- Only in the balanced regime do we obtain the fixed-point survivor  $k_*$  identified in §5.4  $\rightarrow$  Persist.

## 6.3 Survivability Window — [DERIVED]

We can now state the Survivability Window for causal structure in terms of the operator channels:

A universe within the jurisdiction of AoC\_02\_On\_c admits a survivable, observer-independent causal order only if:

1. Constraint is non-zero ( $\zeta > 0$ )

To prevent simultaneity failure. Without Constraint, cones flatten and any ordering is destroyed by boosts and  $\xi$ .

2. Extension is non-zero ( $\omega > 0$ )

To prevent isolation collapse. Without Extension, no non-trivial influence connects distinct events; history cannot propagate.

3. Chance is non-zero but bounded ( $0 < \xi < \xi_{\text{crit}}$ )

To break degeneracy and reject fine-tuned geometries, while remaining small enough that the topology of the light cones and the partial order  $\prec$  survive.

Within this window, the opposing tendencies of  $\zeta$  (which suppresses reach) and  $\omega$  (which promotes reach), under the adjudication of  $\xi$ , drive the recursive dynamics of  $\hat{\Omega}$  toward a fixed-ratio null slope:

- If  $\zeta$  dominates excessively  $\rightarrow$  cones freeze,  $k \rightarrow 0$ .
- If  $\omega$  dominates with insufficient  $\zeta \rightarrow$  cones flatten,  $k \rightarrow \infty$ .
- If  $\xi$  is absent or overwhelming  $\rightarrow$  no stable null-slope class survives.

In the balanced regime, the recursion selects a finite, non-zero, frame-invariant null-slope class  $k_*$  as a structural attractor. AoC\_02 identifies this  $k_*$  as the Invariant Null Slope: the geometric survivor that later maps, via calibration (AoC\_03 and beyond), to the constant we label  $c$ .

Status:

[DERIVED: Within the balanced regime of  $(\zeta, \omega, \xi)$ , a finite, non-zero, frame-invariant null slope  $k_*$  is the unique structural survivor labeled Persist by  $\hat{\Omega}$ .

## 7. Units Firewall & Dimensionless Invariant

In accordance with AoC\_00 §5 (Units Firewall), we strictly separate:

- the structural output of the Collapse Operator (a dimensionless invariant), from
- its metrological mapping (a dimensional quantity expressed in human units).

This prevents the category error of claiming to derive a unit-dependent number from unit-independent geometry.

### 7.1 Primary Survivor — [DERIVED]

The structural survivor identified in §5.4 is the equivalence class of invariant null slopes, denoted  $k_*$ .

By construction,  $k_*$  is:

- finite,
- non-zero, and
- frame-invariant under the admissible transformations (Ansatz\_B).

Because of this, there always exists a preferred class of coordinate charts—Naturalized Charts—in which the null structure takes the form:

$$ds^2 = - dt^2 + dx^2 + dy^2 + dz^2$$

for massless propagation. In such charts, the coordinate speed of null signals satisfies:

$$\left| \frac{dx}{dt} \right| = 1$$

for any spatial direction aligned with the null ray. In other words, there is a normalization in which the dimensionless null slope takes the canonical value:

$$k_* = 1.$$

This “1” is not “1 meter per second” or “1 light-year per year”. It is the pure ratio between spatial and temporal coordinates after we absorb the structural speed limit into our choice of units. Operationally:

- $k_*$  encodes the fixed relation between the Extension channel ( $\omega$ ) and the Constraint channel ( $\zeta$ ) in the survivable regime,
- but its dimensionless value can always be normalized to 1 by rescaling the time and/or length units.

Status:

[DERIVED: A finite, non-zero, frame-invariant null slope  $k_*$  exists, and there is always a normalization where  $k_* = 1$ .]

## 7.2 Separation from Metrology — [CALIBRATION]

The mapping from the dimensionless survivor  $k_*$  to the dimensional constant  $c$  measured in SI units is not a derivation; it is a calibration step that depends on external scale choices. In order to obtain a numerical value like:

$$c = 299,792,458 \frac{\text{m}}{\text{s}},$$

one must introduce independent unit scales. This can happen in (at least) two ways:

### 1. Empirical Definition (current SI practice)

- The SI system defines  $c$  to be exactly 299 792 458 m/s.
- This choice fixes the length unit “meter” in terms of the time unit “second” and the structural relation encoded by  $k_*$ .
- In this view,  $k_* = 1$  and “299 792 458” is a convention, not a prediction.

### 2. Downstream Theoretical Calibration (future AoC entries)

- Future Atlas entries (e.g. AoC\_03 via the fine-structure constant  $\alpha$ ) may relate dimensionless combinations of constants to geometric scales (Bohr radius, Planck units, etc.).
- Even then, once a unit system is chosen (SI, Planck, “atomic” units, alien civilization units), the numerical value of  $c$  follows from that choice of scale, not from  $\hat{\Omega}$  alone.

Thus:

- $k_*$  is a structural output of the operator.
- $c$  (in m/s or any other unit) is a calibrated label attached to that structure.

Status:

[CALIBRATION: The SI value  $c = 299,792,458\text{m/s}$  is a metrological convention, not a prediction of  $\hat{\Omega}$ .]

### 7.3 The Non-Claim Rule — [JURISDICTION]

To keep the Units Firewall airtight, we codify the boundary of what AoC\_02 does and does not claim.

Non-Claims:

1. This entry does not claim to derive the specific numerical value “299 792 458”.
2. This entry does not claim that other civilizations, choosing different base units, would obtain the same number for  $c$  in their native units.
3. This entry does not assert a unique dimensional value of  $c$  independent of unit choice.

Claims (within jurisdiction):

1. Given the preconditions of §3, a finite, non-zero, frame-invariant null-slope structure  $k_*$  must exist (§§5–6).
2. This  $k_*$  is the necessary outcome of the tension between Constraint ( $\zeta$ ), Extension ( $\omega$ ), and Chance ( $\xi$ ) under the Collapse Operator  $\hat{\Omega}$ .
3. In an appropriate naturalized unit system, the dimensionless invariant can always be normalized such that  $k_* = 1$ .

Status:

[JURISDICTION: AoC\_02 derives the existence and properties of the invariant null slope  $k_*$ ; it does not derive any specific unit-dependent value of  $c$ .]

## 8. Exclusion of Competing Geometries

This section provides a schematic moat around the derivation. AoC\_02 has already shown, within its declared domain (§3), that only a finite, non-zero, frame-invariant null slope  $k_*$  returns Persist under  $\hat{\Omega}$  (§5–§6).

Here we briefly examine several popular alternative causal structures and indicate how, when  $\hat{\Omega}$  is formally extended to them, they fail the survivability criteria of §5.1.

### Scope Note:

These alternatives typically violate one or more preconditions of §3 (e.g. Lorentzian signature, Phenomenological Sequence). Strictly speaking, they are out of AoC\_02's jurisdiction. The analysis below is therefore tagged as [SCHEMATIC EXCLUSION]: it is meant to guide future Atlas entries (e.g. “Selection Law”), not to enlarge AoC\_02's formal domain.

### 8.1 Catalog of Alternative Causal Structures — [SCHEMATIC]

We list four broad classes of departures from a single-time, Lorentzian, invariant-cone geometry:

#### 1. Instantaneous Action Geometries (Newton–Cartan-type)

- Structure: Absolute time, absolute simultaneity; interactions propagate with effectively infinite speed ( $k \rightarrow \infty$ ).
- Null structure: Collapses to spatial hypersurfaces of “Now”.

#### 2. Multi-Time Signatures (2T, 3T, ...)

- Structure: Spacetimes with more than one timelike dimension, e.g. signature (2,2) or (2,3).
- Null structure: Cones become “temporal sheets” in which one can turn within a time–time plane.

#### 3. Fundamentally Dispersive Vacuum (Energy-Dependent Cones)

- Structure: The fundamental limiting speed in vacuum depends on the energy of the probe,  $k = k(E)$ , not just on medium effects (this is beyond ordinary plasma dispersion).
- Null structure: Different energies define inequivalent “maximal” cones.



#### 4. Fundamentally Anisotropic Vacuum (Direction-Dependent Cones)

- Structure: The fundamental limiting speed depends on direction,  $k = k(\hat{n})$ ; there is a preferred spatial vector field.
- Null structure: Local cones are squashed or stretched along preferred directions.

Each of these attempts to host “causal structure” without a single, isotropic, frame-invariant null slope.

## 8.2 Application of the $\hat{\Omega}$ Criterion — [EXCLUSION]

We now apply, schematically, the Survivable Sequence criteria (§5.1) and the Viability Map R (§1a.6) to each class. For each, we indicate which condition(s) fail and which  $\hat{\Omega}$  label results.

### 8.2.1 Instantaneous Action (Newton–Cartan-Type, $k \rightarrow \infty$ )

- Structure:
  - Absolute time, absolute simultaneity; signals propagate with effectively infinite speed.
  - “Null surfaces” are constant-time slices; any two spatially separated events at the same absolute time are mutually reachable “instantaneously”.
- Failure Mechanism (under extended  $\hat{\Omega} + \xi$ ):
  1. Simultaneity tipping under frame jitter:

Even a small admixture of space–time mixing in the admissible transformations (the boost-like part of  $\xi$ , §1a.5) tilts the simultaneity hyperplanes.

- In frame  $F$ :  $A$  and  $B$  are simultaneous and mutually connected.
- In nearby frame  $F'$ : the same hyperplane becomes slanted, so either  $A < B$  or  $B < A$ .

## 2. Order inversion:

By chaining such small tilts, one can construct loops where an event lies in its own causal future:  $A < B < A$ .

- Survivability Criteria:
  - Violates Condition 2 (Transformation Robustness): ordering depends on frame.
  - Violates Condition 3 (Perturbation Stability): arbitrarily small  $\xi$ -perturbations scramble simultaneity slices.
- $\hat{\Omega}$  Verdict:

$R \Rightarrow$  Diverge (Simultaneity Failure)

This is the §5.3 “Case A” pathology re-expressed in geometric language.

## 8.2.2 Multi-Time Signatures ( $p \geq 2$ Timelike Dimensions)

- Structure:
  - Signature  $(p, q)$  with  $p \geq 2$ .
  - There exists a time–time plane in which one can steer a curve while remaining locally “timelike”.
- Failure Mechanism (generic):
  1. Closed timelike curves by temporal steering:

With two timelike axes, one can follow a timelike direction along  $t_1$ , then rotate into  $t_2$ , and so on, returning to the starting event while remaining within the “timelike” region at each step.

## 2. Generic cyclicity:

For generic metrics with  $p \geq 2$ , closed timelike curves are not isolated curiosities but structurally allowed; small perturbations of the metric do not remove them.

- Survivability Criteria:

- Violates Condition 1 (Frame-Local Acyclicity):

The relation  $<$  fails irreflexivity; there exist events with  $A < A$  via a closed timelike curve.

- $\xi$ -perturbations do not fix this; they can create additional loops rather than removing them.

- $\hat{\Omega}$  Verdict:

$R \Rightarrow$  Diverge (Cyclicity / Identity Failure)

Multi-time signatures cannot support a Survivable Sequence as defined in §5.1.

### 8.2.3 Fundamentally Dispersive Vacuum (Energy-Dependent Limiting Speed $k(E)$ )

Note:

Ordinary medium dispersion (e.g. light in glass) is not at issue; those cases still have a universal vacuum limit. Here we consider models where the vacuum light cone itself depends on energy.

- Structure:
  - High-energy and low-energy quanta define different null cones.
  - At each point, there is no single maximal slope; instead a family  $k(E)$ .
- Failure Mechanism (schematic loop construction):

1. Energy-stratified cones:

In frame  $F$ , a high-energy signal from  $A$  to  $B$  may be allowed (inside its cone), while the corresponding low-energy signal is not.

2. Boost enhancement:

Under an admissible boost (Ansatz\_B) combined with  $\xi$ , the high-energy signal can be further blue-shifted so that in frame  $F'$  its effective cone changes relative to the low-energy one.

3. Ordering paradox:

By appropriate choice of emission events and frames, one can arrange for a high-energy signal to arrive “before” a low-energy emission in  $F'$ , while preserving the causal ordering in  $F$ . This creates cross-frame contradictions in  $\prec$ .

- Survivability Criteria:
  - Violates Condition 2 (Transformation Robustness):

The classification “inside/outside causal cone” is energy- and frame-dependent; what is causal in one observer’s description can become acausal—or even retro-causal—in another’s.

- Typically violates Condition 3 (Perturbation Stability):  
 $\xi$ -induced frame/energy jitter amplifies these discrepancies.
- $\hat{\Omega}$  Verdict:

$R \Rightarrow$  Diverge (Invariance Instability)

The would-be “speed limit” ceases to be a single survivor; instead, causal structure fractures into incompatible energy layers.

## 8.2.4 Fundamentally Anisotropic Vacuum (Direction-Dependent $k(\hat{n})$ )

Again: small anisotropies in matter distributions are not at issue. We consider theories where the fundamental local null cone of vacuum is direction-dependent.

- Structure:
  - The limiting speed depends on direction  $\hat{n}$ .
  - There exists a vacuum “preferred direction” field  $v^\mu_{\text{pref}}$  that picks out faster vs slower propagation.
- Failure Mechanism (with  $\xi$ ):
  1. Preferred direction vs frame jitter:

Under  $\xi$ ’s frame rotations and boosts (§1a.5), the orientation of  $v^\mu_{\text{pref}}$  relative to local observers changes unpredictably.

## 2. Patchwork cones:

Different regions, or even different realizations of  $\xi$ , will see different “fast directions”. The null-slope class  $K^{(n)}$  becomes patchy and history-dependent.

## 3. Global incoherence:

When attempting to stitch together a global  $\prec$ , segments that rely on different preferred directions conflict. Loops or ill-defined comparabilities appear generically.

- Survivability Criteria:

- At best,  $\xi$  averages the anisotropy out, driving  $k(\hat{n})$  toward an isotropic effective  $k_*$ .
- In that case, the original anisotropic vacuum is not a survivor; it is washed out by Chance.
- At worst, anisotropy plus  $\xi$  leads to inconsistent ordering across regions and frames.
- This violates Condition 2 (Transformation Robustness) and can trigger Condition 1 failures via induced loops.

- $\hat{\Omega}$  Verdict:

$R \Rightarrow$  Diverge or Collapse (Identity Loss or Average-Out)

Either the anisotropy is destroyed by  $\xi$  (reducing back to the isotropic null cone of AoC\_02), or the causal structure fails to define a global Survivable Sequence.

## 8.3 Result — [DERIVED / CONDITIONAL]

Collecting the exclusions:

- Infinite slope ( $k \rightarrow \infty$ , Newton–Cartan-type)  
 $\rightarrow$  Fails Transformation Robustness and Perturbation Stability  $\rightarrow$  Diverge (Simultaneity Failure).
- Multi-time signatures ( $p \geq 2$  timelike)

→ Fails Acyclicity → Diverge (Cyclicity / Identity Failure).

- Fundamentally energy-dependent slope ( $k(E)$ )

→ Fails Transformation Robustness (and typically Perturbation Stability) → Diverge (Invariance Instability).

- Fundamentally direction-dependent slope ( $k(\hat{n})$ )

→ Either washed out by  $\xi$  (original anisotropy not a survivor) or leads to global inconsistencies → Diverge or Collapse.

Within the AoC\_02 preconditions (§3) and the operator semantics of AoC\_00, the only causal geometry that qualifies as a Survivor is:

- Lorentzian signature with a single time, and
- a finite, non-zero, frame-invariant local null cone, corresponding to the invariant null slope  $k_*$  derived in §5.

The “Speed of Light” in this Atlas entry is not an arbitrary decorator constant.

It is the unique null-cone structure that survives the requirement of having a coherent,  $\xi$ -stable history.

Status:

[DERIVED (conditional on §3): Primary competing geometric classes fail the survivability criteria; AoC\_02's invariant null cone stands as the only survivor within this framework.}]

## 9. Falsification & Replication Plan

This section specifies when AoC\_02 must be withdrawn or revised and how an independent verifier can attempt to reproduce its core claim.

In the language of AoC\_00 §7 (Falsifiers) and §9 (Hall of Records / Replication), AoC\_02 asserts:

Given the preconditions of §3, a finite, invariant null-slope structure  $k_*$  is necessary for any universe that supports stable, observer-independent histories.

AoC\_02 is therefore scientific only to the extent that there exist clear observations that would contradict this, and clear protocols that allow others to re-run the argument via independent methods.

### 9.1 Falsification Hooks — FALSIFIER

AoC\_02 is falsified (or forced into major revision) if empirical observation robustly shows that the vacuum null structure is not a single, frame-invariant survivor.

Below, each falsifier is tied explicitly to the Survivable Sequence criteria (§5.1) and the Viability Map (§1a.6).

#### 9.1.1 Frame-Dependent Vacuum Null Slope

- Observable:

High-precision experiments find that different inertial frames (after all known corrections) measure different vacuum null slopes for light in empty space:

$$k_{\text{vac}}(F_1) \neq k_{\text{vac}}(F_2)$$

for admissible frames  $F_1, F_2$  related by an Ansatz\_B transformation.

- Structural Hit:
  - Violates Condition 2 (Transformation Robustness) of §5.1:  
     $\prec$  is no longer invariant under boosts.
  - The null-slope class  $K^{(n)}$  depends on observer;  $\hat{\Omega}$  cannot label a single equivalence class as survivor.

- Section(s) Broken:
  - §5.3 (Case C exclusion)
  - §5.4 (Fixed-point survivor: invariance assumption)
- AoC\_02 Status if observed:

→ Tag FALSIFIED: Boost-dependent vacuum null slope  
 Either AoC\_02's preconditions or its conclusion must be revised.

## 9.1.2 Stable Superluminal Signaling in Vacuum

- Observable:

Repeatable, controllable information transfer in vacuum with signal speed  $v_{\text{signal}} > k_{\text{vac}}$ , without catastrophic paradoxes elsewhere (no global breakdown that would itself falsify the observation).

- Structural Hit:
  - Demonstrates that the Constraint channel  $\zeta$  does not bound causal influence at  $k_*$ .
  - Reverts causal structure toward Case A ( $k \rightarrow \infty$ ) behavior in §4.2 / §5.3.
- Section(s) Broken:
  - §4.2 (Simultaneity vs Sequence: exclusion of unbounded extension)
  - §5.3.1 (Rejection of  $k \rightarrow \infty$ )
- AoC\_02 Status if observed:

→ Tag FALSIFIED: Superluminal vacuum signaling

The claim that a finite upper bound is structurally necessary fails.

---



### 9.1.3 Energy-Dependent Vacuum Dispersion (Fundamental $k(E)$ )

Distinction: Ordinary medium dispersion (glass, plasma) is not a falsifier; AoC\_02 concerns the vacuum null cone itself.

- Observable:

Robust, source-independent evidence that light in vacuum exhibits fundamental energy-dependent speed:

$$k_{\text{vac}}(E_1) \neq k_{\text{vac}}(E_2)$$

for photons (or analogous massless quanta) emitted simultaneously and propagating through empty space.

- Structural Hit:
  - Violates Condition 2 (Transformation Robustness): boosting mixes energies, so what is “inside” the cone at one energy can become “outside” at another.
  - Typically violates Condition 3 (Perturbation Stability):  $\xi$ -induced frame / energy jitter amplifies inconsistency in  $K^{(n)}$ .
  - This is exactly the §5.3 “Case C” failure: variable / drifting slope.
- Section(s) Broken:
  - §5.3.3 (Rejection of variable/drifting slope)
  - §8.2.3 (Exclusion of dispersive vacuum geometries)
- AoC\_02 Status if observed:

→ Tag FALSIFIED: Energy-dependent vacuum null slope

Or, minimally, UNDER REVISION: Ansätze /  $\xi$ -class until the operator is redefined.

---

## 9.1.4 Epoch-Dependent Drift of the Vacuum Null Slope

- Observable:

Long-baseline cosmological measurements (e.g., time-of-flight comparisons across redshift) show that, after accounting for all calibration and cosmological-model effects, the vacuum null slope itself drifts with cosmic time:

$$k_{\text{vac}}(t_1) \neq k_{\text{vac}}(t_2)$$

in a way that cannot be absorbed into changing units or local media effects.

- Structural Hit:
  - Violates Condition 3 (Perturbation Stability):  $k_*$  is no longer a fixed point of  $\hat{\Omega}$ ; the recursive process does not converge.
  - The survivor is replaced by a slowly drifting attractor  $k(t)$ , contradicting §5.4.
- Section(s) Broken:
  - §5.4.3 (Perturbation stability of  $k_*$ )
  - §6.3 (Survivability Window: fixed ratio from  $\zeta/\omega$  balance)
- AoC\_02 Status if observed:

→ Tag FALSIFIED: Null slope not a fixed point

Or reclassify the claim as epoch-local rather than structural.

---

## 9.1.5 Fundamental Vacuum Anisotropy

- Observable:

Convincing evidence (e.g., from modern Michelson–Morley-type, cavity, or astrophysical tests) that vacuum propagation speed depends on direction  $\hat{n}$  in a way not attributable to local media:

$$k_{\text{vac}}(\hat{n}_1) \neq k_{\text{vac}}(\hat{n}_2)$$

with a stable, preferred orientation.

- Structural Hit:
  - Introduces a vacuum preferred direction field; violates Condition 2 (Transformation Robustness) under  $\xi$ -induced rotations.
- Either:
  - $\xi$  washes out anisotropy  $\rightarrow$  the fundamental anisotropic geometry is not a survivor (it collapses back to AoC\_02's isotropic case), or
  - $\xi$  plus anisotropy creates global inconsistencies / loops (per §8.2.4).
- Section(s) Broken:
  - §5.1 (Survivable Sequence: frame robustness)
  - §8.2.4 (Anisotropic vacuum exclusion)
- AoC\_02 Status if observed as truly fundamental:

$\rightarrow$  Tag FALSIFIED: Direction-dependent vacuum null slope

---

## 9.2 Replication Paths — VERIFICATION PROTOCOL

To enter the Hall of Records (AoC\_00 §9) as more than a one-off argument, AoC\_02 must admit independent reconstructions of its core result. We therefore identify two complementary verification paths:

- Path A: Reproduce the result using the Recursive Face of  $\hat{\Omega}$ .
- Path B: Cross-check using the Spectral Face (wave-operator / characteristic analysis).

Success of both strengthens the claim and tests Bridge Claims in AoC\_00; disagreement forces the entry into [UNDERDETERMINED] status.

### 9.2.1 Path A — Recursive Viability Analysis

Tag: VERIFICATION PROTOCOL — Recursive

- Objective:

Starting from a generic initial state  $M_0$  (§5.2) and a declared  $\xi$ -class (§1a.5), demonstrate that recursive application

$$M_{n+1} = \hat{\Omega}(M_n)$$

drives:

- All  $k \rightarrow 0$ ,  $k \rightarrow \infty$ , and  $k$ -variable regimes to Collapse or Diverge, and
- Only a finite, non-zero, frame-invariant class  $k_*$  to Persist.
- Protocol Sketch:
  1. Specify Domain:  
Implement Ansatz\_L, Ansatz\_B, Ansatz\_ $\xi$  exactly as in §1a.2–§1a.5 / §3.
  2. Draw Generic  $M_0$ :  
Choose  $g_{\mu\nu}^{(0)}$  with Lorentzian signature but arbitrarily varying local slopes; ensure no pre-tuning of cones.

3. Instantiate  $\xi$ :

Define a bounded, smooth noise class  $h_{\mu\nu}(\lambda)$  and frame jitter  $\Phi_\lambda$  with documented amplitudes  $\varepsilon_\xi$  and bandwidths.

4. Iterate:

For many seeds  $\lambda$  and initial states  $M_0$ , compute  $M_n$ , extract  $K^{(n)}$  via  $G$ , and evaluate  $R$ .

- Success Metric:

- Across seeds and generic initial conditions, the only long-term behavior flagged Persist is convergence of  $K^{(n)}$  to a single equivalence class  $k_*$  (finite, non-zero, frame-invariant).
- Any other stable attractor (e.g., a nontrivial anisotropic or dispersive fixed point) would contradict §5.3–§5.4 and trigger a re-audit.

## 9.2.2 Path B — Spectral / Wave-Operator Analysis

Tag: VERIFICATION PROTOCOL — Spectral

- Objective:

Use the Spectral Face of  $\hat{\Omega}$  (AoC\_00 §1.2) to show that the same invariant structure  $k_*$  emerges from stability of the wave operator, not just from recursive ordering arguments.

- Protocol Sketch:

1. Construct Wave Operator:

On the Lorentzian manifold ( $g_{\mu\nu}$  in the survivability window), form the d'Alembertian  $\square = g^{\mu\nu} \nabla_\mu \nabla_\nu$  or analogous kinetic operator for massless fields.

2. Analyze Characteristics:

Identify the characteristic surfaces of  $\square$ ; these should coincide with the null cones determined by  $k_*$ .

3. Apply  $\xi$ -Style Perturbations:

Perturb  $g_{\mu\nu} \rightarrow g_{\mu\nu} + \varepsilon_\xi h_{\mu\nu}$  and examine:

- Stability of the characteristic set (do null characteristics remain sharply defined?),
- Whether any alternative characteristic speeds appear as robust modes.

#### 4. Bridge Check:

Confirm that the only dispersionless,  $\xi$ -stable characteristic structure matches the invariant null slope obtained via Path A.

- Success Metric:
  - The principal symbol of the wave operator continues to single out a unique characteristic speed under perturbation.
  - No secondary characteristic speed appears as  $\xi$ -stable.
  - If the spectral analysis supports a different structure than the recursive face (e.g., multiple  $\xi$ -stable characteristic cones), then Bridge Claim(s) in AoC\_00 and the status of AoC\_02 both become [UNDERDETERMINED].

### 9.3 Hall of Records & Status Tags — RECORD-KEEPING

For the Atlas archivist ( $\hat{\Omega}$ 's Hall of Records), AoC\_02 must carry dynamic status tags based on empirical and theoretical developments:

- VALID (PROVISIONAL)
  - No falsifier in §9.1 has been observed.
  - Path A completed; Path B pending or partially complete.
- VALID (CROSS-VERIFIED)
  - No falsifier observed.
  - Both Path A and Path B reproduce the same invariant structure  $k_*$  within declared tolerances.
- FALSIFIED

- Any falsifier in §9.1 is robustly confirmed and cannot be absorbed by revising preconditions in §3 without contradiction.
- AoC\_02 remains in the record only as a historical or limiting-case entry.
- UNDERDETERMINED
  - Recursive and spectral paths disagree on the survivor structure, or
  - Empirical data strongly suggest tension (e.g., hints of dispersion / anisotropy) but are not yet decisive.
- OUT OF JURISDICTION
  - Future Atlas entries may demonstrate that our universe does not satisfy AoC\_02's preconditions (e.g., fails Phenomenological Sequence, or operates on a fundamentally different signature).
  - In that case AoC\_02 is not "wrong" but explicitly marked as not applicable to the realized branch.

STATUS (AT TIME OF WRITING THIS ENTRY):

[VALID (PROVISIONAL): No known falsifier decisively contradicts a single, invariant vacuum null slope; cross-verification is an open program.]

## 10. Downstream Dependencies

This section documents AoC\_02 as load-bearing infrastructure within the Atlas.

Because the Collapse Operator  $\hat{\Omega}$  builds structure recursively, the stability of the survivor null-slope invariant  $k_*$  is a precondition for later entries that describe interaction strengths, bound states, and long-range forces.

Units Firewall reminder. In the Atlas,  $k_*$  denotes the dimensionless survivor class of null slopes (the invariant causal structure). The symbol  $c$  denotes a metrological representative (a calibrated numerical value in a chosen unit system) associated with that invariant. The physics claim is about  $k_*$ ; the SI number attached to  $c$  is [CALIBRATION]. (This matches the firewall language already present elsewhere in the entry.)

If AoC\_02 is falsified under the criteria of §9, every downstream entry that assumes a finite, frame-invariant  $k_*$  must be re-audited.

### 10.1 Role in the Atlas Dependency Spine — INFRASTRUCTURE

In the Prooffield Index, the core sequence of geometric/causal kernels is:

- AoC\_00 (Collapse Operator  $\hat{\Omega}$ ) — Selection Kernel:

Defines the recursive viability machinery (Persist/Collapse/Diverge,  $\xi$ -class, Units Firewall).

- AoC\_01 (On  $\pi$ ) — Geometric Kernel:

Establishes Spatial Closure and the existence of a stable manifold that can host propagation.

- AoC\_02 (On  $c/k_*$ ) — Causal Kernel:

Selects the survivor null slope  $k_*$  and the associated invariant causal structure.

Downstream entries — including at minimum AoC\_03 ( $\alpha$ ), AoC\_04 ( $m_p$ ), AoC\_05 ( $m_n$ ), and AoC\_06 ( $G$ ) — all assume that:

1. The universe admits a single, finite, non-zero, frame-invariant survivor null slope  $k_*$ , and



2. This structure behaves as a fixed point of  $\hat{\Omega}$  within the survivability window (§6.3).

### Cascade Consequence.

If AoC\_02 is tagged [FALSIFIED] or [OUT OF JURISDICTION], then AoC\_03–AoC\_06 must be flagged [CONTINGENT — RE-AUDIT REQUIRED] until they are either:

- Re-derived under a revised causal kernel, or
- Explicitly restricted to a regime where AoC\_02 still applies (e.g., low-energy, late-epoch limit).

## 10.2 AoC\_03: Fine-Structure Constant $\alpha$ — CONTINGENT

Role of  $c$  (as calibrated representative of  $k_*$ ).

The fine-structure constant encodes the strength of electromagnetic interaction. In standard form,

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c},$$

so the denominator explicitly contains  $c$ . At the structural level,  $\alpha$  measures “coupling strength per causal interaction event,” which presupposes:

1. A well-defined, invariant causal structure (a single survivor  $k_*$ ), and
2. A stable distinction between “before” and “after” for emission/absorption processes.

### How AoC\_03 breaks if AoC\_02 fails.

- If  $c$  (hence the null slope) is frame-dependent: different inertial observers assign different  $\alpha$ , collapsing universality.
- If  $c$  is energy-dependent (fundamental vacuum dispersion):  $\alpha$  becomes process-dependent already at tree level, i.e.  $\alpha = \alpha(E)$ .
- If  $c \rightarrow \infty$  (unbounded extension): a localized “interaction event” becomes ill-posed; simultaneity erases the time structure underlying “per-event” coupling.

Status: [CONTINGENT ON AoC\_02] — Any derivation of  $\alpha$  in AoC\_03 assumes a finite, frame-invariant survivor  $k_*$  (and its calibrated representative  $c$ ).

### 10.3 AoC\_04 & AoC\_05: Nucleon Masses ( $m_p, m_n$ ) — CONTINGENT

Role of  $c/k_*$ .

The proton and neutron masses are dominated by QCD binding energy, not bare quark masses. Confinement relies on:

1. Finite-speed gauge propagation (gluons respecting the same null structure that defines  $k_*$ ), and
2. A causal confinement scale (order  $\sim 1$  fm) that functions as an effective horizon for colored degrees of freedom.

Within the Atlas', AoC\_04–AoC\_05 (On  $m_p$  and  $m_n$  respectively) we implicitly assume that QCD dynamics occur on top of the AoC\_02 causal background.

How AoC\_04–AoC\_05 break if AoC\_02 fails.

- If  $c \rightarrow 0$  (zero extension / no propagation): there is no causal channel for gluon exchange; binding cannot be mediated; confinement (and thus nucleon mass as we know it) fails.
- If  $c \rightarrow \infty$  (unbounded extension / instantaneous propagation): retardation and dynamical confinement lose meaning; the usual Hamiltonian/QFT constructions yielding stable  $m_p$  and  $m_n$  become structurally ill-posed.
- If  $c$  is energy-dependent: the effective confinement scale becomes probe-dependent; “mass” varies with measurement energy, contradicting universal rest mass.

Status: [CONTINGENT ON AoC\_02] — nucleon mass entries assume finite, non-zero, invariant propagation set by  $k_*$  (with  $c$  as its calibration).

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## 10.4 AoC\_06: Gravitational Constant $G$ — CONTINGENT

Role of  $c/k_*$ .

In metric gravity theories (including GR and any AoC-compatible successors), gravitational influence propagates along or within the same null structure that defines  $k_*$ . The constant  $G$  is meaningful only in a context where:

1. Gravitational disturbances propagate at a definite speed (gravitational waves), and
2. Causal horizons (black-hole event horizons, cosmological horizons) are tied to a finite null slope.

How AoC\_06 breaks if AoC\_02 fails.

- If  $c \rightarrow \infty$  (instantaneous gravity): gravity collapses into action-at-a-distance; gravitational waves and horizon structures disappear; relativistic  $G$  tied to curvature propagation loses meaning.
- If  $c \rightarrow 0$  (no gravitational propagation): curvature becomes rigid/static; any derivation relying on propagation, retardation, or radiation balance is invalidated.
- If  $c$  is frame-dependent: observers measure incompatible gravitational signal speeds and thus incompatible effective  $G$ , undermining the Equivalence Principle and any Atlas entry treating  $G$  as a scalar constant.

Status: [CONTINGENT ON AoC\_02] — long-range gravitational coherence and the operational meaning of  $G$  assume an invariant null slope  $k_*$ .

## 10.5 Broader Atlas Impact — SCHEMATIC

Beyond AoC\_03–AoC\_06, the blast radius of an AoC\_02 failure extends across the Atlas:

- Electromagnetic Scales: the Bohr radius  $a_0$ , the Rydberg constant  $R_\infty$ , and related atomic scales depend on  $c$  through  $\alpha$ . Any violation of AoC\_02 propagates into these derived quantities via AoC\_03.
- Cosmology: causal horizons ( $ct$ ), the particle horizon, and estimates of the age and size of the universe use  $c$  as the null slope defining “how far

influence can travel in time  $t$ .” If  $c$  drifts or fragments, the cosmological distance ladder must be rebuilt.

- Standard Model Couplings and  $RG$  Flow:  $RG$  running is defined on a background with well-defined energy–momentum relations and lightcone structure. If the vacuum null slope is not invariant, the usual identification of scales, thresholds, and “high-energy limits” must be revisited.

Status: [INFRASTRUCTURE] — AoC\_02 provides the causal floor on which high-energy physics and cosmology are built. A change of status for AoC\_02 triggers a Cascade Re-Audit for all entries that assume an invariant  $k_*$  (and its calibrated representative  $c$ ).

## 11. Heading for Section 11

This section defines the audit protocol for an external verifier (human or AGI). The goal is to check if this entry's claims follow rigorously from its explicit preconditions and the Atlas framework (AoC\_00, AoC\_01).

A verifier must treat this as a structured inspection of:

- Declared domain and ansätze
- Correct wiring of the Collapse Operator
- Logical exclusion of alternatives
- Proper Units Firewall discipline
- Falsifiability and replication hooks
- Documented downstream dependencies

Only if all structural checks pass may the entry retain its [DERIVED] status under AoC\_00.

### 11.1 Pre-Flight: Ansätze & Domain

Goal: Confirm that all load-bearing premises are surfaced and correctly tagged, and that no hidden domain assumptions are smuggled in.

#### Domain & Prior Entries

- AoC\_00 dependency: The entry explicitly cites the Collapse Operator  $\hat{\Omega}$  (recursive/viability face, Persist/Collapse/Diverge, noise class  $\xi$ , Units Firewall) as [DERIVED] from AoC\_00. (§3.1, §1a.1, §1a.5, §1a.6)
- AoC\_01 dependency: Spatial Closure ( $\pi$ ) and the existence of a stable manifold are taken as prior results, not re-derived. (§3.1)

#### Ansätze & Phenomenology

- Ansatz\_L (Lorentzian): The Lorentzian signature  $(-+++)$  and global hyperbolicity are explicitly declared as [ANSATZ], not portrayed as a result of AoC\_02. (§1a.2, §3.2.1)

- Ansatz\_B (Boost family): The existence of a non-trivial class of admissible “boost-like” frame transformations is explicitly tagged as [ANSATZ]. (§3.2.2)
- Phenomenological Sequence Requirement: The lived existence of stable “before/after” histories is clearly tagged as [PHENOMENOLOGICAL REQUIREMENT], and used to rule out frozen/simultaneous universes. (§3.2.3)
- Ansatz\_ξ (Noise class): The noise field ξ is declared non-zero, bounded, and reproducible (parameters λ, amplitude ε\_ξ, distribution class), and tagged as [ANSATZ]. (§1a.5, §3.2.4)

FAIL condition: If any of these premises is used but not explicitly declared and tagged, the entry is at least [UNDERDETERMINED] for domain ambiguity.

## 11.2 Operator Wiring: State, Extractor, Viability Map

Goal: Confirm that the selection is genuinely done by  $\hat{\Omega}+\xi+R$ , not by informal argument or hidden assumptions.

### State & Extractor:

- State definition: The system state at recursion depth is defined as

$$M_n = (g_{\mu\nu}^{(n)}, C^{(n)}),$$

where is a Lorentzian metric candidate and the associated causal structure (future-directed light cones / partial order). (§1a.3)

- Extractor definition: The geometry extractor is explicitly defined as

$$G(M_n) = (N^{(n)}, K^{(n)}),$$

where is the null-structure field and the null-slope class, constructed from coordinate slopes of null directions. (§1a.4)

### Noise & Viability Map:

- Noise implementation: ξ is defined as a mapping

$$\xi(\lambda) : M_n \mapsto M_n(\lambda),$$

with two components: metric fluctuations and frame jitter via admissible  $\Phi_\lambda$ , with  $\lambda$  including seeds and amplitudes. (§1a.5)

- Viability Map R: The map

$$R(\{G(M_n(\lambda))\}) \in \{\text{Persist, Collapse, Diverge}\}$$

is explicitly defined, with operational criteria for each outcome in terms of (null structure, null-slope class, and causal order). (§1a.6)

[PASS requirement: A verifier must be able to reconstruct, from the text alone, the full operator pipeline]

$$M_n \xrightarrow{\xi} M_n(\lambda) \xrightarrow{G} (N^{(n)}(\lambda), K^{(n)}(\lambda)) \xrightarrow{R} \{\text{Persist, Collapse, Diverge}\}.$$

### 11.3 Core Derivation: Excluding Pathologies, Identifying the Survivor

Goal: Verify that all non-survivor regimes are excluded by the same survivability criteria, and that the survivor class is uniquely characterized.

#### Survivable Sequence Definition

- Three conditions present: The definition of a Survivable Sequence (partial order  $<$  on events ) includes all three:
  1. Frame-local acyclicity (strict partial order in each admissible frame).
  2. Transformation robustness (no order inversion under admissible boosts).
  3. Perturbation stability (no order inversion under  $\xi$ -perturbations). (§5.1)

#### Exclusion of Pathological Regimes (Case A/B/C)

- Case A —  $k \rightarrow \infty$  (Unbounded extension): The text shows that effectively instantaneous propagation produces frame-dependent simultaneity slices that tilt under  $\xi$  (boost jitter), generating order inversions and/or loops  $\rightarrow$  violation of Conditions 2 and 3  $\rightarrow$  Diverge. (§5.3.1)
- Case B —  $k \rightarrow 0$  (Zero extension): The text shows that with zero propagation, events cannot influence spatial neighbors; the universe fractures into isolated

world-lines, violating the Phenomenological Sequence Requirement (no non-trivial  $<$  connecting distinct spatial points)  $\rightarrow$  Collapse. (§5.3.2)

- Case C —  $k$  variable (frame/position/energy dependent): The text shows how differing null slopes between frames or energies allow construction of causal loops, violating Condition 2 (and often 3)  $\rightarrow$  Diverge. (§5.3.3)

### Survivor Verification

- Candidate  $k_*$  defined: The survivor candidate is explicitly described as a null-slope class with properties: finite, non-zero, frame-invariant. (§5.4 opening)
- Condition 1 (Acyclicity) checked: Finite prevents cone flattening; no frame's causal structure yields closed causal loops  $\rightarrow$  transitivity + irreflexivity hold. (§5.4.1)
- Condition 2 (Transformation robustness) checked: Invariance of under admissible boosts ensures that the causal classification (timelike/null/spacelike) of separations — and hence the order  $<$  — cannot be inverted by frame change. (§5.4.2)
- Condition 3 (Perturbation stability) checked: With finite and non-zero, light-cone “fuzzing” under  $\xi$  leaves a finite safety margin; the partial order's topology (no loops, no collapse) is preserved under small bounded perturbations. (§5.4.3)
- Uniqueness claim scoped: The entry explicitly states that within the declared preconditions (§3), regimes  $k \rightarrow \infty$ ,  $k \rightarrow 0$ , and  $k$ -variable all map to Collapse/Diverge, leaving only the finite, non-zero, invariant class as Persist. (§5.4.4–5)

[FAIL condition: If any pathological regime is dismissed without reference to the same survivability criteria (Conditions 1–3 +  $\xi$  + R), the exclusion is incomplete.]

## 11.4 Stress Tests & Survivability Window

Goal: Ensure that the  $\zeta/\omega/\xi$  interpretation in §6 actually matches the behavior derived in §5 and is not decorative.

- Channel mapping: The roles of  $\zeta$  (Constraint / locality),  $\omega$  (Extension / reach), and  $\xi$  (Chance / noise) are clearly mapped to: speed limit, drive to connect, and perturbation testing. (§6.1)
- Limit table coherence: The perturbation table in §6.2 correctly matches limits of  $\zeta$ ,  $\omega$ ,  $\xi$  to earlier behavior:



- $\zeta \rightarrow 0 \leftrightarrow \text{Case A (unbounded extension)} \rightarrow \text{Diverge}$
- $\omega \rightarrow 0 \leftrightarrow \text{Case B (zero reach)} \rightarrow \text{Collapse}$
- $\xi \rightarrow 0 \text{ or } \xi \rightarrow \infty \leftrightarrow \text{no selection or noise domination} \rightarrow \text{Diverge}$
- Balanced  $\zeta, \omega, \xi \leftrightarrow \text{survivor window} \rightarrow \text{Persist}$
- Survivability window stated: §6.3 explicitly defines the region where  $\zeta, \omega, \xi$  are all non-zero and bounded, and identifies this as the only regime in which a finite, invariant survives.

## 11.5 Firewall, Exclusions, and Dependencies

Goal: Confirm that the entry does not overclaim (Units Firewall), that competing geometries are treated honestly, and that downstream dependencies are documented, not hidden.

### Units Firewall & Dimensionless Invariant:

- Primary survivor is dimensionless: The survivor is identified as a dimensionless equivalence class of null slopes, and a natural normalization where is explicitly stated. (§7.1)
- Metrology separated: The mapping  $\rightarrow$  (e.g. 299,792,458 m/s) is explicitly tagged as [CALIBRATION], and the text states that this depends on the chosen unit system, not on  $\hat{\Omega}$ . (§7.2)
- Non-claim rule: The entry explicitly disclaims deriving the SI numerical value of, and clarifies that different civilizations could have chosen different labels while sharing the same structural invariant. (§7.3)

### Competitive Geometries (Section 8)

- Alternatives catalogued: The four schematic competitor classes (Newton–Cartan, multi-time, dispersive vacuum, anisotropic vacuum) are listed with clear structural descriptions. (§8.1)
- $\hat{\Omega}$ -based rejection: Each alternative is rejected by explicit reference to how it violates the survivability criteria (acyclicity, frame robustness, perturbation stability) and/or the preconditions (§3), and is tagged as schematic, not fully rigorous. (§8.2, §8.3)

## Downstream Dependencies (Section 10)

- Kernel role stated: AoC\_02 is identified as the Causal Kernel in the Atlas spine  $\text{AoC\_00} \rightarrow \text{AoC\_01} \rightarrow \text{AoC\_02}$ , with a documented cascade re-audit requirement if AoC\_02 fails. (§10.1)
- AoC\_03–AoC\_06 contingent: The dependencies of AoC\_03 ( $\alpha$ ), AoC\_04–05 ( $m_p$ ,  $m_n$ ), and AoC\_06 ( $G$ ) on an invariant are clearly stated, with [CONTINGENT ON AoC\_02] tags. (§10.2–§10.4)
- Broader impact noted: The schematic impact on EM scales, cosmology, and SM couplings is listed and tagged as [SCHEMATIC] or [INFRASTRUCTURE], not overclaimed as derived. (§10.5)

## 11.6 Falsifiability & Replication

Goal: Confirm that the entry is empirically vulnerable and reproducible, in line with AoC\_00 §7–§9.

### Falsifiers:

- Multiple hooks: At least three explicit falsification hooks (frame dependence, superluminal signaling, energy dependence, epoch drift, anisotropy) are listed. (§9.1)
- Section-level hits: Each hook is tied to specific structural failures: which Survivable Sequence condition and which section of the derivation break if the observation is confirmed. (§9.1)

### Replication:

- Path A — Recursive: A concrete recursive protocol (generic  $\xi$ , iteration of  $\hat{\Omega}$ , measurement of  $K^{(n)}$  and  $R$ ) is provided as a verification path. (§9.2, Path A)
- Path B — Spectral: A cross-check via the spectral/wave-operator face (characteristics of  $\square$ ,  $\xi$ -perturbations, characteristic speed matching) is specified to test the Bridge Claim between recursive and spectral faces. (§9.2, Path B)

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## 11.7 Verifier Verdict

After completing the checklist, the verifier must assign one of the following AoC\_02 status tags (see §9.3):

### VALID (PROVISIONAL)

- All structural and firewall items above are checked; no known falsifier is confirmed. Path A is at least conceptually implementable; Path B may be pending.

### VALID (CROSS-VERIFIED)

- As above, and Path A + Path B have both been executed (analytically or numerically) and found to agree on the same invariant .

### FALSIFIED

- Any falsification hook from §9.1 is robustly observed in experiment, or the derivation demonstrably violates its own survivability criteria.

### UNDERDETERMINED

- One or more critical checklist items (especially in §§11.1–11.3, 11.5) cannot be confirmed, or Path A and Path B disagree on the survivor structure.

### OUT OF JURISDICTION

- The universe under consideration does not satisfy the preconditions (§3), e.g., it does not admit a Lorentzian structure or lived phenomenological sequence.

Instruction to AUPEI Verifier:

Record the final status tag in the Hall of Records, along with a brief note citing which checklist items failed (if any), and which (if any) falsification hooks have been triggered.

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